WasteWatch

Automated Illegal-Dumping Detection & Alert System

Technical & Operational Briefing

Prepared for review by municipal / regulatory authority

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1. Purpose and Scope

WasteWatch is a camera-based monitoring system designed to detect and document instances of illegal waste dumping in public and municipal areas. It is intended to support enforcement teams by automatically identifying suspicious activity, verifying that activity using artificial intelligence, and presenting confirmed cases to a human reviewer before any formal record or alert is issued.

This document explains, in non-technical terms, how the system observes an area, how it decides something may be a violation, how it double-checks that decision using AI, and — most importantly — where and how a human being remains in control of every consequential decision the system makes.

The system is not designed to autonomously issue fines, penalties, or legal notices. Its role is limited to detection, evidence capture, and alerting; all enforcement action remains the responsibility of the receiving municipal authority or field personnel.

2. Plain-Language Overview of How the System Works

The system operates in four broad stages, illustrated below:

Stage	What happens
1. Watch	Cameras installed at monitored locations continuously stream video into the system.
2. Detect	Software identifies people and vehicles in the video and tracks how long they linger in a defined restricted zone (for example, an area next to a public bin where dumping is prohibited).
3. Verify	When lingering behaviour crosses a time threshold, a snapshot is sent to an AI visual-analysis model, which is asked to describe what it sees and judge whether the behaviour looks like dumping.
4. Alert & Review	If the AI flags the behaviour as likely dumping, an alert is raised to the operations dashboard and, for higher-severity cases, to a phone/SMS escalation channel. A human reviewer can inspect the video evidence and confirm or reject the finding.

> No single stage acts alone. A camera seeing a person stand still is not, by itself, a violation. A violation record is only created after the movement pattern, the AI's visual judgement, and (where applicable) a human reviewer's confirmation all agree.

3. Detection Stage — What the Cameras and Software Actually See

Video from each camera is processed by an object-detection model that identifies people and vehicles (the system does not attempt to identify specific individuals by face, and does not perform facial recognition). Each detected person or vehicle is assigned a temporary tracking number for as long as they remain visible in the frame — this number is not linked to any personal identity and is discarded once the person leaves the camera's view.

The software watches only for movement within pre-configured "restricted zones" drawn onto each camera's view by the municipal operator — for example, the area immediately around a public waste bin, or a stretch of roadside where dumping has previously occurred. Activity outside these zones is not evaluated for violations.

Time-Based Escalation

A person or vehicle entering a restricted zone does not immediately trigger anything. The system watches how long they remain and how they behave:

- **Brief presence** (e.g., someone walking through or past the zone) — no action taken, no record kept.
- **Extended presence** (default: more than 5 seconds) — the system begins closer observation.
- **Prolonged presence or a suspicious motion pattern** (default: more than 15 seconds, or a motion consistent with placing/throwing an object) — the system escalates to AI verification.

These time thresholds are configurable per site by the municipal operator, based on local conditions (e.g., a bus stop within a restricted zone would need a longer threshold than an isolated alley).

4. Verification Stage — Why a Second AI Check Exists Before Any Alert

Movement and dwell-time alone are not reliable evidence of wrongdoing — a person could be waiting for transport, resting, or attending to a vehicle. To avoid raising false alarms, the system does not treat prolonged presence as a confirmed violation. Instead, it captures a still image of the moment and a short surrounding context image, and submits both to a Vision-Language Model (an AI system trained to interpret and describe images in natural language).

The AI model is asked a structured set of questions equivalent to:

> "Describe what this person/vehicle is doing. Does this look like waste is being dumped or discarded? If so, what type of waste? How confident are you? How severe does this appear?"

The model responds with a structured judgement — a description of the action, a yes/no assessment of whether dumping appears to be occurring, an estimated waste category, a confidence score, and a severity rating. This response is what actually determines whether an alert is raised — the camera detection alone never does.

Model Used

The system currently uses a vision-language model (Qwen2.5-VL) hosted entirely on infrastructure controlled by the operator — no images are sent to any third-party cloud AI service. A planned upgrade will move this analysis to a newer model (MiniCPM-o 4.5), also hosted on the same self-controlled infrastructure, chosen for improved accuracy and the ability to additionally provide continuous scene narration (see Section 10).

> All AI processing happens on hardware located at, or under the direct control of, the operating authority. No camera footage or derived images leave the system's own servers as part of this analysis step.

5. Human Oversight — Nothing Becomes an Official Violation Without Review

This is the most important safeguard in the system's design. The AI's judgement is treated as a recommendation, not a verdict:

- Every AI-flagged event is logged with its supporting evidence — the triggering image(s), the AI's stated reasoning, its confidence score, and the camera/location/time.
- A designated operator or supervisor can review this evidence through a case-history dashboard before any action beyond an internal alert is taken.
- Under the planned continuous-narration enhancement (Section 10), an additional category of AI observation is deliberately routed into a **separate, lower-confidence review queue** rather than being treated as a confirmed violation at all — a human must explicitly confirm it before it is even recorded as an official violation event.

- The system logs who reviewed each case, what decision was made, and when — creating an accountable record of human decision-making, not just machine output.

> **Design safeguard:** The system's default behaviour is designed so that AI uncertainty results in a human being asked to look, not in an automatic penalty being issued. Where the AI's judgement is less structured or less certain (free-form observation rather than a confirmed detection), the system requires human confirmation before that observation is elevated to the same status as a fully verified violation.

6. Alerting and Escalation Process

Once a violation is confirmed (either directly by the AI-verification step, or after human confirmation of a narration-derived observation), the system distributes notifications through the following channels, depending on configured severity:

Channel	Used for
Operations dashboard (live view)	All confirmed events, viewable in real time by monitoring staff
Messaging notification (e.g., Telegram)	Immediate notification to designated staff/officers
Automated phone call escalation	Higher-severity or repeat-offence cases, where immediate attention is warranted
On-site audio announcement	Optional real-time deterrent message played at the camera location itself

Repeat activity at the same location by the same tracked individual within a session (recognised only by the temporary tracking number, not personal identity) can escalate the severity and prompt the phone-call channel, ensuring persistent issues receive prompt human attention.

7. Evidence Handling, Retention, and Auditability

For every confirmed or under-review event, the system automatically retains:

- A snapshot image at the moment of detection
- A short video clip covering the period immediately before and after the event
- The AI's structured analysis and reasoning text
- Camera identity, timestamp, and location zone
- The review outcome and the identity of the reviewing staff member, once reviewed

Evidence is stored on the operator's own servers, with an automated backup copy synchronised to a secure cloud storage location for redundancy. Each record is timestamped and immutable once written, supporting its use as an auditable record should enforcement action or a dispute follow.

8. Privacy and Data Protection Safeguards

- **No facial recognition.** The system identifies "a person" or "a vehicle," not who that person is. It does not match faces against any identity database.
- **Temporary tracking only.** The numeric tracking ID assigned to a person exists only while they remain visible on camera and is not linked across separate visits or to any external identity system.
- **Zone-limited monitoring.** Only pre-defined restricted zones are evaluated for violations; general pedestrian movement outside those zones is not flagged or stored as a case.

- **Self-hosted AI processing.** All AI analysis of imagery occurs on infrastructure controlled by the operating authority — imagery is not transmitted to third-party commercial AI providers.
- **Access-controlled evidence.** Review of stored evidence requires authenticated staff access, with role-based permissions (viewer / administrator) governing who can view versus who can act on a case.

9. System Reliability and Monitoring

The system includes automated health monitoring of every camera feed and internal service. If a camera goes offline, loses signal, or a processing component fails, this is detected automatically and, where possible, the affected component is automatically restarted. Persistent failures are surfaced to operations staff so that a gap in monitoring coverage is never silent.

10. Planned Enhancements

An enhancement currently in design (not yet in operational use) will add a **continuous scene-narration capability** — rather than only reacting when a person lingers in a zone, the AI will maintain an ongoing, real-time description of activity in each camera's view. This is intended to catch violation types that a movement/dwell-time detector alone cannot, such as:

- The act of throwing or depositing waste outside a designated bin, distinct from mere presence near it
- A vehicle passing through with an uncovered or overflowing load

As described in Section 5, any observation from this continuous-narration capability is, by design, routed to a dedicated human-review queue rather than being auto-recorded as a confirmed violation — reflecting the lower certainty inherent in open-ended AI observation compared to the structured, threshold-triggered verification process described in Sections 3–4.

11. Glossary of Terms

Term	Plain-language meaning
Restricted zone	An area drawn on a camera's view where dumping monitoring is active (e.g., around a public bin)
Dwell time	How long a detected person or vehicle remains in a restricted zone
Vision-Language Model (VLM)	An AI system that can look at an image and describe/interpret it in written language
Confidence score	The AI's own stated certainty in its judgement, expressed as a percentage
Tracking ID	A temporary number assigned to a detected person/vehicle for as long as they remain visible; not a personal identity
Escalation	Raising the urgency/notification channel of an alert (e.g., from dashboard-only to a phone call) based on severity or repeat activity
Human review queue	A list of AI-flagged observations awaiting confirmation or dismissal by an authorised staff member before becoming an official record

This document describes the system's design and operating principles as of August 2026 for the purpose of regulatory and municipal review. Technical implementation details are available on request for independent audit.